

ORF307: Optimization

Precept: #3

Agenda

- Flop count
- Least squares
- Positive (Semi-)Definite Matrices and Gram matrices
- Least squares interpretation
- More coding practice (see notebook in Companion).

Reminders and Announcements

- Always check Python notebooks in Companion (in directories `lectures/` and `precepts/`)
- Write down the deadline of the current HW:
- Other announcements on blackboard.

Flop count

Consider $A, B \in \mathbb{R}^{m \times n}$, $C \in \mathbb{R}^{n \times p}$, $x \in \mathbb{R}^n$, and $\alpha \in \mathbb{R}$

Matrix operation	Flop count
αA	mn
$A + B$	mn
$x^T x$	$2n - 1$
Ax	$(2n - 1)m$
AC	$(2n - 1)mp$
$A^T A$	$\approx mnp$

- Vector multiplication flops:

$$x^T x = \sum_{i=1}^n x_i^2$$

n multiplications and $n - 1$ additions

- Matrix-vector multiplication flops:

$$Ax = \begin{bmatrix} \text{---} & \mathbf{a}_1^T & \text{---} \\ \text{---} & \mathbf{a}_2^T & \text{---} \\ & \vdots & \\ \text{---} & \mathbf{a}_m^T & \text{---} \end{bmatrix} x = \begin{bmatrix} \mathbf{a}_1^T x \\ \mathbf{a}_2^T x \\ \vdots \\ \mathbf{a}_m^T x \end{bmatrix}$$

We have m dot products so we have $(2n - 1)m$ flops.

- Matrix multiplication

$$AC = A \begin{bmatrix} | & | & & | \\ \mathbf{c}^1 & \mathbf{c}^2 & \dots & \mathbf{c}^p \\ | & | & & | \end{bmatrix} = \begin{bmatrix} | & | & & | \\ \mathbf{Ac}^1 & \mathbf{Ac}^2 & \dots & \mathbf{Ac}^p \\ | & | & & | \end{bmatrix}$$

We have p matrix-vector products so we have $(2n - 1)mp$ flops.

- For the matrix product $A^T A$ we have one-half of the flops because the matrix $A^T A$ is symmetric so we only need to compute around half of the flops of the general case (i.e., $(1/2)(2n - 1)mp \approx nmp$)

Least squares

Consider the optimization problem

$$\min_x \|Ax - b\|_2^2$$

where the matrix $A \in \mathbb{R}^{m \times n}$ is tall (which means we have many more rows than columns and the system is **overdetermined**).

NOTE: When a number m is much larger than another number n we can use the notation: $m \gg n$.

Derive the normal equations:

The objective function can be written as

$$\|Ax - b\|_2^2 = (Ax - b)^T (Ax - b) = x^T A^T Ax - 2b^T Ax + b^T b$$

We define the First Order Necessary Conditions (FONC) for optimality

$$\nabla \|Ax - b\|_2^2 = 0 \implies 2A^T Ax - 2A^T b = 0$$

The normal equations are therefore defined as the system of linear equations

$$A^T Ax = A^T b$$

and its solution x^* is the solution of the least squares problem.

Positive (semi) definite matrices

Definition: A matrix $A \in \mathbb{R}^{n \times n}$ is called positive semidefinite matrix (PSD) iff

$$x^T A x \geq 0, \forall x \in \mathbb{R}^n$$

Notation: $A \succeq 0$

Definition: A matrix $A \in \mathbb{R}^{n \times n}$ is called positive definite matrix (PD) iff

$$x^T A x > 0, \forall x \in \mathbb{R}^n, x \neq 0$$

Gram matrices:

The matrix $A^T A$ is called the **Gram matrix** of A .

Show that $A^T A$ has the following properties:

- $A^T A$ is symmetric matrix.

Proof: A matrix X is symmetric iff $X^T = X$. We also know that $(XY)^T = Y^T X^T$. Then we have:

$$(A^T A)^T = A^T (A^T)^T = A^T A$$

- $A^T A \succ 0$ (i.e., $A^T A$ is Positive Definite (PD) matrix) if the columns of A are linearly independent.

Proof: Recall that a matrix H is PD **iff** $x^T H x > 0, \forall x \in \mathbb{R}^n$ and $x \neq 0$.

Now take any nonzero n -dimensional vector x and:

$$x^T A^T A x = (Ax)^T A x = \|Ax\|_2^2$$

Since the columns of A are linear independent we must have $Ax = 0$ only when $x = 0$. Hence $\|Ax\|_2^2 > 0, \forall x \neq 0$, and the matrix $A^T A$ is PD (or you can write equivalently $A^T A \succ 0$).

Least squares interpretation

Interpret the least squares problem and its solution using the following example:

$$\min \|Ax - b\|_2^2$$

where

$$A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix}, \quad b = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$$

The aim of this problem is to find a point x^* such that the distance (i.e., the $L2$ -norm) between the vectors Ax^* and b is minimized.

The vector Ax lives in the column space of the matrix A , which is the $x_1 x_2$ -plane in $\mathbb{R}^3$. You can easily see this by computing the product: $Ax = (x_1, x_2, 0)$. On the other

hand, the vector $b = (1, 1, 1)^T$ is not on the x_1x_2 -plane. Hence, the least squares problem is to find a point x^* in the x_1x_2 -plane such that Ax^* is closest to b .

The solution of this least squares problem is $x^* = (1, 1)^T$ lives in $\mathbb{R}^2$. However, the vector $Ax^* = (1, 1, 0)^T$ lives in $\mathbb{R}^3$.

For this particular example, Ax^* is the projection of point b to the x_1x_2 -plane (see Figure 1).

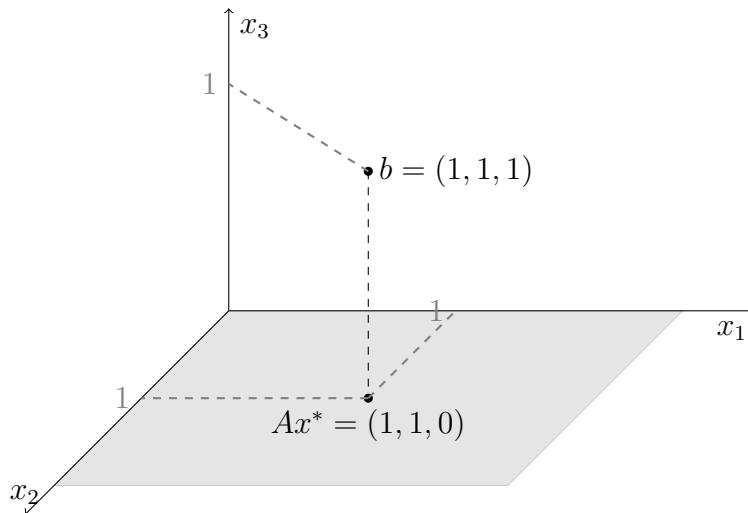


Figure 1: Interpretation of least squares. For this particular example (selection of A and b) the solution of the optimization problem gives us the projection of b onto the x_1x_2 -plane.

Least squares orthogonality principle:

Prove that the residual vector $r = Ax^* - b$ is orthogonal to any vector $Ax, \forall x \in \mathbb{R}^n$. In other words, we always have:

$$Ax \perp r, \forall x \in \mathbb{R}^n$$

Proof:

Using the definition of orthogonal vectors we need to show that:

$$(Ax)^T r = 0, \forall x \in \mathbb{R}^n$$

. This is easy to show. Consider the normal equations:

$$\begin{aligned} A^T Ax^* = A^T b &\implies A^T (Ax^* - b) = 0 \implies A^T r = 0 \implies r^T A = 0 \implies \\ &\implies r^T Ax = 0 \implies (Ax)^T r = 0 \implies Ax \perp r, \forall x \in \mathbb{R}^n \end{aligned}$$

In the previous example the residual vector is

$$r = Ax^* - b = (1, 1, 0)^T - (1, 1, 1)^T = (0, 0, -1)^T$$

It is easy to verify that this vector is orthogonal to any vector $Ax, \forall x \in \mathbb{R}^2$.

Notice also that r is also perpendicular to both columns of A . Hence every linear combination of those columns (Ax) must also be perpendicular (orthogonal) to r .